Supplementary Material: Segment–Task Specialization Does Not Emerge in Active-Dendrite Networks

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Abstract

Supplementary material for *Segment–Task Specialization Does Not Emerge in Active-Dendrite Networks: A Dead-Neuron Artifact, a Repair That Does Not Transfer, and What the Own-Segment Bias Actually Does*. This document supplies the per-phase records behind the signed-drift measurement, a line-by-line reconciliation of the pre-registered criteria against the executed grids, and the full equivalence-testing results for every comparison the main text describes as showing no detectable difference.

1 S1. Per-phase signed-drift records

The main text reports the signed-drift measurement in summary (one-sample t over seeds per phase, and the per-phase mean displacement). This appendix gives the full per-phase records: the own-gate displacement d_{own} (the change in the own task's column of the dendritic gate over the phase), the column norm, the raw own-gate magnitude at the phase start and end, and the active-only own-segment rate.

2 S2. Pre-registered criteria against the executed grid

The three studies in this manuscript each carried a pre-registration written before data collection. Where the executed grid differs from the pre-registered denominator, the deviation is listed here rather than absorbed. Every item below was disclosed in the main text at the point where it is used.

3 S3. Equivalence testing

The main text reports several comparisons as "no difference detected" or "reproduces the base system". A failure to reject is not evidence of equivalence, so each such comparison is accompanied here by a two-one-sided-tests (TOST) equivalence test at a pre-declared margin of equivalence of 0.5 accuracy points, reported as a 90% confidence interval (the interval that corresponds to $\alpha = 0.05$ in a TOST). A comparison is declared equivalent only when the whole 90% CI lies inside $[-0.5, 0.5]$; otherwise the honest statement is that a difference of up to the CI half-width cannot be excluded at $n = 10$.

Two of these results deserve to be stated plainly rather than left to the reader. First, the granularity ladder's null is *not* uniformly supported by equivalence testing: the branch-versus-neuron comparison on the learning term is equivalent at the declared margin in the two MNIST cells but not in the Fashion cells and not for final accuracy, so the correct summary is that we detected no difference larger than roughly half a point in the cells where the effect is small, not that the two constructions are interchangeable everywhere. Second, the constant-gate control does not formally reproduce the base system: its 90% CI on the learning term is $[-1.58, +0.92]$ on Fashion-MNIST and $[-0.43, +0.57]$ on MNIST, so an effect of up to about one and a half points cannot be excluded on Fashion-MNIST. The main text therefore describes the constant-gate arm as being consistent with the base system rather than as reproducing it.

4 S4. Additional controls: full statistics

The main text reports the additional controls added after the first review of this line of work in summary. This appendix gives the underlying statistics for each of them: the own-segment specificity arms, the positive control for the dominant-segment instrument, the anchor-construction variants, and the comparison with contemporary replay methods. Every comparison is a paired t-test over $n = 10$ seeds sharing permutations and initialization; p_{Holm} is the Holm-corrected value within each metric family, d_z the paired standardized effect size, and the interval is the 95% CI for the paired mean difference.

5 S5. Task-relevance probe: per-group diagnostics

The task-relevance probe of the main text, with the per-group per-task diagnostic accuracy for each dose. Ten interleaved phases: two high-relevance (in-domain), two medium-relevance (rotated count), and six zero-relevance tasks. Half of the zero-relevance tasks use labels rotated within each class across tasks, so that the class structure stays learnable and only the task–label mapping changes; the other half use fully random labels, whose accuracy is chance by construction.

6 S6. Attribution of the mechanism measurements: training, initialization, or the gradient mask

The three quantities measured in the mechanism section — the winning-segment gate value, the own-segment win rate, and the prediction flip rate — are reported here for three states of the network: never trained (*fresh*), trained with the per-task gradient mask as in the paper (*protected*), and trained with the mask switched off (*unmasked*). Means over $n = 3$ seeds per dataset. The gate value and the own-segment win rate are already at their final values in the untrained network; the flip rate is the only one of the three that training changes materially.

Table 1: Per-phase signed-drift records. d_{own} is the displacement of the own taskś dendritic-gate column over the phase; u_{own} is the raw (bias-free) own-gate magnitude; p_{own} is the active-only own-segment rate read from the bias-free instrument. Three seeds per arm.

seed	b	phase	d_{own}	u_{own} (end)	$\|W_{\text{col}}\|$	p_{own}
0	0	0	+0.0042	-0.078	9.259	0.110
0	0	1	+0.0007	-0.017	9.380	0.044
0	0	2	+0.0010	+0.123	9.255	0.111
0	0	3	+0.0020	+0.283	9.367	0.131
0	0	4	+0.0005	+0.315	9.359	0.058
0	0	5	-0.0001	-0.215	9.153	0.075
0	0	6	+0.0008	-0.217	9.184	0.065
0	0	7	+0.0008	-0.199	9.195	0.093
0	0	8	-0.0007	-0.048	9.164	0.107
0	0	9	-0.0001	-0.164	9.344	0.078
1	0	0	+0.0044	+0.231	9.225	0.120
1	0	1	+0.0015	-0.092	9.261	0.091
1	0	2	+0.0014	-0.107	9.302	0.094
1	0	3	-0.0003	+0.253	9.110	0.057
1	0	4	+0.0011	-0.189	9.197	0.117
1	0	5	+0.0011	+0.302	9.196	0.103
1	0	6	+0.0014	-0.276	9.201	0.101
1	0	7	+0.0020	-0.055	9.353	0.123
1	0	8	-0.0010	-0.089	9.105	0.070
1	0	9	+0.0021	-0.166	9.225	0.128
2	0	0	+0.0031	-0.121	9.248	0.084
2	0	1	+0.0011	+0.102	9.325	0.061
2	0	2	-0.0000	+0.266	9.292	0.030
2	0	3	-0.0002	-0.108	9.269	0.046
2	0	4	-0.0003	+0.230	9.308	0.135
2	0	5	-0.0004	-0.249	9.315	0.086
2	0	6	+0.0006	+0.301	9.178	0.064
2	0	7	-0.0001	-0.059	9.266	0.036
2	0	8	-0.0001	+0.279	9.341	0.059
2	0	9	+0.0010	-0.047	9.255	0.071
0	10	0	+0.0270	-0.076	9.237	0.159
0	10	1	+0.0147	-0.032	9.367	0.083
0	10	2	+0.0075	+0.123	9.252	0.128
0	10	3	+0.0067	+0.280	9.366	0.193
0	10	4	+0.0068	+0.315	9.362	0.084
0	10	5	+0.0046	-0.192	9.175	0.127
0	10	6	+0.0033	-0.217	9.173	0.097
0	10	7	+0.0050	-0.167	9.209	0.154
0	10	8	+0.0036	-0.042	9.176	0.133
0	10	9	+0.0024	-0.126	9.362	0.118
1	10	0	+0.0268	+0.270	9.213	0.152
1	10	1	+0.0141	-0.092	9.252	0.116
1	10	2	+0.0091	-0.071	9.300	0.154
1	10	3	+0.0079	+0.253	9.132	0.168
1	10	4	+0.0079	-0.188	9.200	0.175
1	10	5	+0.0053	+0.302	9.200	0.111
1	10	6	+0.0049	-0.276	9.188	0.115
1	10	7	+0.0039	-0.021	9.361	0.160
1	10	8	+0.0022	-0.089	9.116	0.116
1	10	9	+0.0034	-0.194	9.225	0.127
2	10	0	+0.0266	-0.122	9.232	0.116
2	10	1	+0.0155	+0.103	9.324	0.143
2	10	2	+0.0090	+0.266	9.294	0.071
2	10	3	+0.0082	-0.059	9.285	0.203
2	10	4	+0.0051	+0.230	9.300	0.130
2	10	5	+0.0039	-0.249	9.319	0.137
2	10	6	+0.0041	+0.296	9.168	0.105
2	10	7	+0.0025	-0.065	9.280	0.065
2	10	8	+0.0051	+0.248	9.350	0.138

Table 2: Pre-registered criterion, executed grid, and status for each pre-registered experiment. "Deviation marks a case where the executed grid or the evaluated metric differs from the pre-registered wording; the dated amendment documenting each deviation is in the artifact.

Experiment	Pre-registered criterion	Executed grid	Status
S1 budget (G0)	$\geq$ 4/6 cells with significant positive $\Delta\ell$	8 cells ($\{\text{dataset}\} \times \{M\} \times \{T\}$)	**Deviation:** the denominator was 6 in the pre-registration and 8 in execution; the outcome is reported against the executed grid.
S3 transfer (G1)	$\geq$ 15/20 cells with significant positive $\Delta\ell$; $\geq$ 18/20 sign-consistent; Δavg and $\Delta\ell$ simultaneously significant in $\geq$ 12/20	12 cells ($\{\text{dataset}\} \times \{T\} \times \{b\}$); no saturated cells	**Deviation:** all three thresholds carry the pre-registered 20 denominator and are unreachable at $n = 12$; the thresholds were not rescaled, and the outcome is unchanged under any rescaling.
S3 artifact (G2)	an untrained probe whose naive share exceeds the active share by ≥ 0.05 *and* a trained $b = 0$ arm whose naive share falls back within 0.05 of the active share	the trained-arm half only	**Partial:** the untrained-probe half was not run; the executed half is met trivially. Flagged, not dropped.
S3 mechanism (G3)	conditional on G1 surviving	never triggered	Not applicable: G1 failed.
S3 mortality (E4)	confirms if GN mortality $\leq$ 0.10 *and* the repair vanishes; refutes if mortality $>$ 0.30 or gain retention $\geq$ 50%	80 runs (GN-MLP) + 320-run scan	Met: mortality 0.765/0.855 (refutes the dying-gate hypothesis) with gain retention 17%/26%.
S4 context noise (E1)	at $\rho = 0.1$ the accuracy loss is $\leq$ 2 points and at $\rho = 0.2 \leq$ 5, *and* the per-seed direction agrees across both datasets in $\geq$ 8/10 seeds	$\rho \in \{0, 0.1, 0.2, 0.4\}$, both datasets, $n = 10$	Amplitude conditions met with margin (largest movement 0.25 points). **Deviation:** the per-seed direction condition was not met in strict form (mixed signs, every per-seed shift below 0.18 points).
S4 repair survival (E2)	at $\rho = 0.1$ the paired $\Delta\bar{a} > 0$ and $\geq$ 50% of the $\rho = 0$ gain is retained	as above, crossed with $b \in \{0, 10\}$	**Deviation:** the criterion was written for $\Delta\ell$, but the noise harness records only final-phase per-task accuracies, so it is evaluated on $\Delta\bar{a}$ — the same net-benefit quantity in the form the runs support.

Table 3: TOST equivalence tests at a margin of equivalence of 0.5 points ($n = 10$ paired seeds). p_{TOST} is the two-one-sided-test p-value; a "yes marks comparisons whose 90% CI lies entirely inside the equivalence margin.

comparison	dataset	metric	Δ	90% CI	p_{TOST}	equiv.
branch - neuron	fashion	dell	-0.17	$[-0.50, +0.17]$	0.051	no
branch - neuron	fashion	davg	+0.22	$[-0.27, +0.72]$	0.165	no
branch - neuron	fashion	dbwt	+0.43	$[-0.25, +1.12]$	0.430	no
branch - neuron	fashion	dell	+0.44	$[-0.11, +0.99]$	0.428	no
branch - neuron	fashion	davg	+0.14	$[-0.62, +0.89]$	0.201	no
branch - neuron	fashion	dbwt	-0.33	$[-1.17, +0.51]$	0.360	no
branch - neuron	mnist	dell	-0.07	$[-0.30, +0.17]$	0.004	**yes**
branch - neuron	mnist	davg	+0.07	$[-0.37, +0.51]$	0.054	no
branch - neuron	mnist	dbwt	+0.15	$[-0.13, +0.43]$	0.025	**yes**
plain - neuron	mnist	dell	-7.46	$[-7.86, -7.06]$	1.000	no
plain - neuron	mnist	davg	-6.14	$[-6.66, -5.62]$	1.000	no
plain - neuron	mnist	dbwt	+1.46	$[+1.07, +1.86]$	0.999	no
branch - neuron	mnist	dell	-0.11	$[-0.36, +0.15]$	0.010	**yes**
branch - neuron	mnist	davg	+0.16	$[-0.29, +0.61]$	0.099	no
branch - neuron	mnist	dbwt	+0.28	$[-0.06, +0.63]$	0.139	no
plain - neuron	mnist	dell	-21.95	$[-22.49, -21.41]$	1.000	no
plain - neuron	mnist	davg	-12.25	$[-13.24, -11.26]$	1.000	no
plain - neuron	mnist	dbwt	+10.20	$[+8.87, +11.54]$	1.000	no
const	fashion	dell	-0.33	$[-1.58, +0.92]$	0.405	no
const	fashion	davg	+0.49	$[-0.75, +1.74]$	0.497	no
const	fashion	dbwt	+0.91	$[-0.69, +2.52]$	0.676	no
repaired	fashion	dell	+1.39	$[+0.34, +2.43]$	0.922	no
repaired	fashion	davg	+0.69	$[-0.17, +1.55]$	0.654	no
repaired	fashion	dbwt	-0.77	$[-2.04, +0.50]$	0.649	no
nogate	fashion	dell	+1.02	$[-0.18, +2.23]$	0.777	no
nogate	fashion	davg	+2.07	$[+0.52, +3.61]$	0.952	no
nogate	fashion	dbwt	+1.16	$[-0.50, +2.82]$	0.757	no
const	mnist	dell	+0.07	$[-0.43, +0.57]$	0.075	no
const	mnist	davg	-0.25	$[-0.93, +0.44]$	0.259	no
const	mnist	dbwt	-0.35	$[-0.83, +0.13]$	0.295	no
repaired	mnist	dell	+3.32	$[+2.90, +3.73]$	1.000	no
repaired	mnist	davg	+4.80	$[+4.29, +5.32]$	1.000	no
repaired	mnist	dbwt	+1.65	$[+1.20, +2.10]$	0.999	no
nogate	mnist	dell	+2.61	$[+1.99, +3.22]$	1.000	no
nogate	mnist	davg	+3.64	$[+3.06, +4.22]$	1.000	no
nogate	mnist	dbwt	+1.15	$[+0.90, +1.40]$	1.000	no

Table 4: Additional-control statistics, part 1. See the main text for the design of each comparison.

—	—	—	—	—	—	—	—	—	—
fashion	repaired	bwt	10	-0.20	-0.3	1.000	-0.08	[-1.89, +1.49]	no
fashion	rescale	bwt	10	+0.84	+0.8	1.000	+0.26	[-1.45, +3.13]	no
fashion	uniform	bwt	10	-0.16	-0.2	1.000	-0.08	[-1.65, +1.32]	no
fashion	random	ell	10	-0.21	-0.9	0.406	-0.28	[-0.74, +0.33]	no
fashion	repaired	ell	10	+0.93	+1.5	0.316	+0.49	[-0.44, +2.29]	no
fashion	rescale	ell	10	+4.68	+10.1	0.000	+3.18	[+3.63, +5.73]	no
fashion	uniform	ell	10	+1.40	+2.8	0.065	+0.88	[+0.26, +2.54]	no
fashion	random	final_avg	10	-0.02	-0.0	0.966	-0.01	[-1.00, +0.96]	no
fashion	repaired	final_avg	10	+0.75	+1.5	0.320	+0.48	[-0.36, +1.87]	no
fashion	rescale	final_avg	10	+5.43	+6.9	0.000	+2.19	[+3.66, +7.21]	no
fashion	uniform	final_avg	10	+1.25	+3.0	0.045	+0.95	[+0.31, +2.20]	no
mnist	random	bwt	10	+0.44	+4.7	0.001	+1.48	[+0.23, +0.65]	no
mnist	repaired	bwt	10	+1.76	+8.9	0.000	+2.82	[+1.31, +2.21]	no
mnist	rescale	bwt	10	+3.29	+16.0	0.000	+5.05	[+2.82, +3.75]	no
mnist	uniform	bwt	10	+1.62	+7.0	0.000	+2.22	[+1.10, +2.14]	no
mnist	random	ell	10	-0.19	-1.0	0.336	-0.32	[-0.60, +0.23]	no
mnist	repaired	ell	10	+3.14	+13.1	0.000	+4.14	[+2.60, +3.68]	no
mnist	rescale	ell	10	+6.85	+24.5	0.000	+7.74	[+6.21, +7.48]	no
mnist	uniform	ell	10	+3.00	+11.2	0.000	+3.54	[+2.39, +3.60]	no
mnist	random	final_avg	10	+0.21	+1.0	0.354	+0.31	[-0.28, +0.70]	no
mnist	repaired	final_avg	10	+4.72	+18.1	0.000	+5.71	[+4.13, +5.31]	no
mnist	rescale	final_avg	10	+9.81	+32.7	0.000	+10.35	[+9.13, +10.48]	no
mnist	uniform	final_avg	10	+4.46	+13.7	0.000	+4.35	[+3.72, +5.19]	no

Table 5: Additional-control statistics, part 2. See the main text for the design of each comparison.

—	—	—	—	—	—	—
fashion	base	final_avg	-0.75	-1.5	0.160	[-1.66, +0.15]
fashion	random	ell	-1.13	-2.2	0.059	[-2.10, -0.17]
fashion	random	final_avg	-0.77	-1.3	0.230	[-1.87, +0.33]
fashion	rescale	ell	+3.75	+7.2	0.000	[+2.79, +4.70]
fashion	rescale	final_avg	+4.68	+4.8	0.001	[+2.88, +6.48]
fashion	uniform	ell	+0.47	+1.3	0.224	[-0.19, +1.13]
fashion	uniform	final_avg	+0.50	+1.0	0.350	[-0.43, +1.43]
mnist	base	ell	-3.14	-13.1	0.000	[-3.58, -2.70]
mnist	base	final_avg	-4.72	-18.1	0.000	[-5.20, -4.24]
mnist	random	ell	-3.32	-10.7	0.000	[-3.89, -2.75]
mnist	random	final_avg	-4.51	-11.5	0.000	[-5.23, -3.79]
mnist	rescale	ell	+3.71	+19.6	0.000	[+3.36, +4.06]
mnist	rescale	final_avg	+5.08	+18.6	0.000	[+4.58, +5.59]
mnist	uniform	ell	-0.14	-1.3	0.238	[-0.34, +0.06]
mnist	uniform	final_avg	-0.26	-1.3	0.227	[-0.64, +0.11]

Table 6: Additional-control statistics, part 3. See the main text for the design of each comparison.

—	—	—	—	—	—	—	—
fashion	oracle	10	1.0000	[0.964, 1.000]	106	58.00	43.90
fashion	weak_oracle	10	1.0000	[0.961, 1.000]	99	56.92	42.66
mnist	base	10	0.0551	[0.005, 0.188]	107	74.38	63.83
mnist	oracle	10	1.0000	[0.967, 1.000]	119	77.33	68.19
mnist	weak_oracle	10	1.0000	[0.965, 1.000]	113	75.98	66.18

Table 7: Additional-control statistics, part 4. See the main text for the design of each comparison.

—	—	—	—	—	—	—	—	—
fashion	weak_oracle	bwt	10	+0.17	+0.3	1.000	+0.09	[-1.21, +1.55]
fashion	oracle	ell	10	+1.17	+2.5	0.069	+0.79	[+0.11, +2.23]
fashion	weak_oracle	ell	10	+0.10	+0.3	0.804	+0.08	[-0.75, +0.94]
fashion	oracle	final_avg	10	+1.49	+2.0	0.154	+0.63	[-0.20, +3.19]
fashion	weak_oracle	final_avg	10	+0.25	+0.8	0.442	+0.25	[-0.46, +0.96]
mnist	oracle	bwt	10	+1.57	+10.0	0.000	+3.16	[+1.21, +1.92]
mnist	weak_oracle	bwt	10	+0.83	+7.8	0.000	+2.48	[+0.59, +1.08]
mnist	oracle	ell	10	+2.95	+11.5	0.000	+3.63	[+2.37, +3.54]
mnist	weak_oracle	ell	10	+1.60	+7.0	0.000	+2.22	[+1.08, +2.12]
mnist	oracle	final_avg	10	+4.37	+13.9	0.000	+4.41	[+3.66, +5.08]
mnist	weak_oracle	final_avg	10	+2.35	+11.1	0.000	+3.51	[+1.87, +2.83]

Table 8: Additional-control statistics, part 5. See the main text for the design of each comparison.

—	—	—	—	—	—	—	—	—	—	—
fashion	branch_v2	ell	10	58.65	32.96	+1.82	+4.6	0.001	[+1.09, +2.55]	
fashion	branch_v3	ell	10	58.65	32.96	+1.82	+4.6	0.001	[+1.09, +2.55]	
fashion	branch_v1	avg	10	56.79	41.68	-0.73	-1.3	0.234	[-1.77, +0.32]	
fashion	branch_v2	avg	10	58.65	32.96	-9.45	-17.4	0.000	[-10.44, -8.46]	
fashion	branch_v3	avg	10	58.65	32.96	-9.45	-17.4	0.000	[-10.44, -8.46]	
mnist	branch_v1	ell	10	74.42	64.07	+0.05	+0.3	0.757	[-0.22, +0.31]	
mnist	branch_v2	ell	10	76.87	44.10	+2.49	+14.3	0.000	[+2.17, +2.81]	
mnist	branch_v3	ell	10	76.87	44.10	+2.49	+14.3	0.000	[+2.17, +2.81]	
mnist	branch_v1	avg	10	74.42	64.07	+0.25	+1.4	0.193	[-0.07, +0.57]	
mnist	branch_v2	avg	10	76.87	44.10	-19.72	-25.8	0.000	[-21.12, -18.32]	
mnist	branch_v3	avg	10	76.87	44.10	-19.72	-25.8	0.000	[-21.12, -18.32]	

Table 9: Task-relevance probe, per-group mean per-task diagnostic accuracy (%), by bias dose. The gap between the high- and medium-relevance groups narrows monotonically with dose while the zero-relevance group stays at chance.

b	n	high	medium	zero	gap (high−medium)
0	10	94.4	91.4	10.5	+3.0
1	10	94.8	92.7	10.7	+2.1
3	10	95.3	93.5	10.8	+1.7
10	10	95.3	93.8	10.8	+1.6

Table 10: Attribution of the mechanism measurements. The gate value and the own-segment win rate are essentially fixed by the initialization; the flip rate is the only quantity that training moves. Removing the gradient mask frees the off-diagonal blocks but does not produce task–segment specialization.

dataset	arm	gate	own win	flip (mean)	flip (max)
mnist	fresh (untrained)	0.5641	0.1004	0.0445	0.0760
mnist	protected (paper)	0.5646	0.0880	0.1477	0.2407
mnist	unmasked (control)	0.5695	0.0995	0.1837	0.3484
fashion	fresh (untrained)	0.5641	0.1004	0.0560	0.0891
fashion	protected (paper)	0.5644	0.0913	0.2836	0.4976
fashion	unmasked (control)	0.5670	0.0997	0.3404	0.5751

Table 11: Weight displacement by block. Relative L2 displacement from the network own initialization, $n = 3$ seeds per dataset. With the per-task gradient mask on, every off-diagonal block is frozen at exactly its initial value.

dataset	arm	off-diag frozen	off-diag move	diag move
mnist	protected (paper)	100%	0.0000	0.0972
mnist	unmasked (control)	0%	0.1091	0.1092
fashion	protected (paper)	100%	0.0000	0.0697
fashion	unmasked (control)	0%	0.0870	0.0834